Probability Theory Chapter IV

4.1 Technical Terms:

When an experiment is repeated under identical and homogenous conditions, there are two situations

i) Results of the experiment is unique i.e. we get a single outcome of the experiment. Such experiment is known as **Deterministic or Predictable** phenomena.

For example

v = u + at, where u is initial velocity, a is acceleration, t is time and v is final velocity.

$V \propto \frac{1}{P}$, where V is volume and P is pressure.

ii) Result of the experiment is not unique but may results in two or more than two possible outcomes. Such experiment is known as **Probabilistic** phenomena.

For example: A coin is tossed, we are not sure whether head or tail will appear.

A die is thrown, any number from 1 to 6 may appear.

- **Trial:** Suppose an experiment is repeated under identical and homogeneous conditions does not give unique result, but may results in one of several possible outcome. The experiment is known as trial.

  Example : A coin is tossed, a die is thrown, a card is drawn from a pack of card, a candidate appear for interview, a seed is sown.

- **Event:** Outcome of a trial is known as event (or cases).

  Example: A coin is tossed, event are head or tail.

  A die is thrown, events are 1, 2, 3, 4, 5, 6.

  A seed is kept for germination, events are seed will germinate or not germinate.

- **Exhaustive Event:** Total number of possible outcome of any trial is known as exhaustive event.

  Example: A coin is tossed, there are two exhaustive events: H, T

  Two coins are tossed, there are 4 exhaustive events: HH HT TH TT

  Three coins are tossed, there are 8 exhaustive events:

  HHH HHT HTH HTT THH THT TTH TTT

In general if n coins are tossed, exhaustive events are 2^n.

A die is thrown, there are 6 exhaustive events 1, 2, 3, 4, 5, 6

Two dice are thrown, there are 36 exhaustive events

1,1	1,2	1,3	1,4	1,5	1,6
2,1	2,2	2,3	2,4	2,5	2,6
3,1	3,2	3,3	3,4	3,5	3,6
4,1	4,2	4,3	4,4	4,5	4,6
5,1	5,2	5,3	5,4	5,5	5,6
6,1	6,2	6,3	6,4	6,5	6,6

In general if n dice are thrown, exhaustive events are 6^n.

- **Mutually Exclusive events:** Events are mutually exclusive if happening of any event exclude the happening of all other events i.e. no two or more than two events occur simultaneously in the same trial.

  Example: A coin is tossed, event H and T are mutually exclusive

  A die is thrown, events 1, 2, 3, 4, 5 and 6 are mutually exclusive.

- **Favourable Event:** Number of events favourable to an event in a trial is number of outcomes which is responsible for happening of that event.

  Example: Three coins are tossed, there are 8 exhaustive events

  HHH HHT HTH HTT THH THT TTH TTT

  Number of events favourable for exactly one head is 3 (HTT, THT, TTH)

  Number of events favourable for atmost one head is 4 (HTT, THT, TTH, TTT)

  Number of events favourable for atleast one head is 7

  (HHH, HHT, HTH, HTT, THH, THT, TTH)

  Number of events favourable for four head is 0

  Number of events favourable for no head is 1 (TTT)

  Example: Two dice are thrown, there are 36 exhaustive events

1,1	1,2	1,3	1,4	1,5	1,6
2,1	2,2	2,3	2,4	2,5	2,6
3,1	3,2	3,3	3,4	3,5	3,6
4,1	4,2	4,3	4,4	4,5	4,6
5,1	5,2	5,3	5,4	5,5	5,6
6,1	6,2	6,3	6,4	6,5	6,6

Number of events favourable for getting sum more than 9 is 6 i.e. {(4,6), (5,5), (5,6), (6,4), (6,5), (6,6)}

- **Equally likely events:** Outcome of a trial are equally likely if taking into consideration all the events, there is no reason to expect any one event in preference to other event i.e all the events have equal chance to happen.

  Example : A coin is tossed, head and tail are equally likely

  A die is thrown, all the event 1, 2, 3, 4, 5, 6 are equally likely.
- **Independent Event:** If happening of any event is not affected by happening of all remaining event.

4.2 Definition of probability:

4.2.1 Mathematical or classical Definition

4.2.2 Statistical or Empirical Definition

4.2.3 Axiomatic Definition

4.2.1: Mathematical or classical definition of probability: If a trial results in **'n'** exhaustive, mutually exclusive and equally likely events, out of which **'m'** are favourable to the happening of an event E, probability of happening of an event E is ratio of number of favourable event to the exhaustive events.

$$p = \frac{\text{Number of favourable events}}{\text{Exhaustive events}}$$

$$p = \frac{m}{n}$$

Number of event favourable for non-happening of an event = $n - m$

Probability of non happening of an event E is q

$$q = \frac{n-m}{n} = 1 - \frac{m}{n} = 1 - p \text{ or } p + q = 1 \text{ where } 0 \le p \le 1, 0 \le q \le 1$$

Probability p of happening of an event is also known as probability of success and probability q of non-happening of an event is known as probability of failure. If p = 0, it is known as impossible event and p =1 is known as sure (or certain) event.

Limitations:

i) If various outcome of the trial are not equally likely

ii) If exhaustive event of a trial are very large.

4.3 Operations on sets:

- **Sets:** A set is well defined collection of all distinct objects.

  S = {1, 2, 3, 4, 5, 6, 7, 8, 9, 10)

- **Union of two sets A and B:** This set consists of all those elements which belong to either A or B or both. It is denoted by $A \cup B$
- **Intersection of two sets A and B**: This set consists of those elements which belong to both A and B. It is denoted by $A \cap B$.
- **Complement of a Set A:** Set of all those values which do not belong to A. It is denoted by $\overline{A}$

Suppose S = {1, 2, 3, 4, 5, 6, 7, 8, 9, 10}

A = {1, 3, 4, 7, 9}

B = {2, 4, 5, 6, 10}

$(A \cup B)$ = {1, 2, 3, 4, 5, 6, 7, 9, 10}

$(A \cap B)$ ={4}

$\overline{A}$ = {2, 5, 6, 8, 10}

$\overline{B}$ = {1, 3, 7, 8, 9}

Some results

i) $\overline{\overline{A}} = A$

$\overline{\overline{A}}$ = {1, 3, 4, 7, 9} = A

ii) $\overline{(A \cup B)} = \overline{A} \cap \overline{B}$

$(A \cup B)$ = {1, 2, 3, 4, 5, 6, 7, 9, 10}

$\overline{(A \cup B)}$ = 8,

$\overline{A} \cap \overline{B}$ = 8

iii) $\overline{(A \cap B)} = \overline{A} \cup \overline{B}$

$\overline{(A \cap B)}$ = {1, 2, 3, 5, 6, 7, 8, 9, 10}

$\overline{A} \cup \overline{B}$ = {1, 2, 3, 5, 6, 7, 8, 9, 10}

If A and B are any two events

i)	Atleast one of the event occurs	$A \cup B$
ii)	Both event will occur	$A \cap B$.
iii)	None will occur	$\overline{A} \cap \overline{B}$
iv)	Exactly one of them will occur	$(A \cap \overline{B}) \cup (\overline{A} \cap B)$
v)	Event A occurs and B does not	$A \cap \overline{B}$
vi)	If event A and B are mutually exclusive or disjoint events	$A \cap B = \emptyset$

4.4 Theorem:

i) **Probability of an impossible event is 0.**

Proof: $S \cup \emptyset = S$

$P(S \cup \emptyset) = P(S)$

Since S and $\emptyset$ are disjoint events

$P(S) + P(\emptyset) = P(S)$

$P(\emptyset) = P(S) - P(S) = 0$

ii) **Probability of complementary event $\bar{A}$ of event A is $P(\bar{A}) = 1 - P(A)$**

Proof: $A \cup \bar{A} = S$

$P(A \cup \bar{A}) = P(S)$

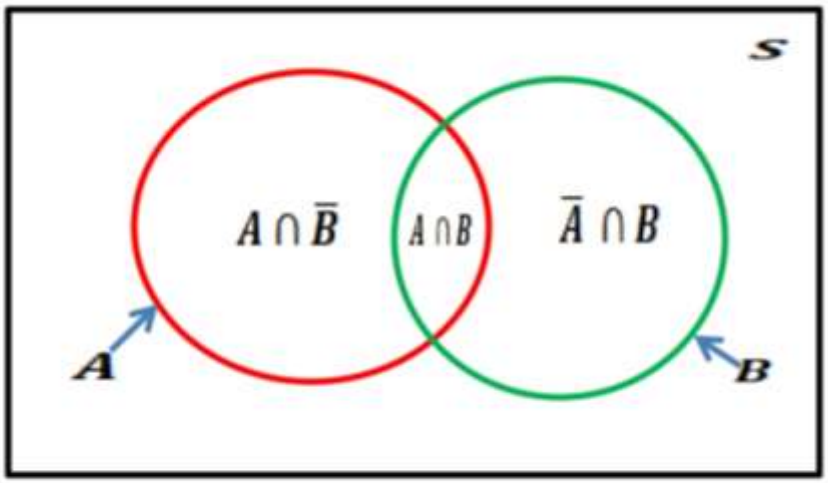
Since A and $\bar{A}$ are disjoint events and $P(S) = 1$

$P(A) + P(\bar{A}) = 1$

$P(\bar{A}) = 1 - P(A)$

iii) **Addition law of probability: If A and B are any two events**

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$



Venn Diagram

Proof: $A \cup B = A \cup (\bar{A} \cap B)$

$P(A \cup B) = P(A \cup (\bar{A} \cap B))$

Since A and $\bar{A} \cap B$ are disjoint event

$P(A \cup B) = P(A) + P(\bar{A} \cap B)$

Add and subtract $P(A \cap B)$

$P(A \cup B) = P(A) + P(\bar{A} \cap B) + P(A \cap B) - P(A \cap B)$

Now $(\bar{A} \cap B)$ and $(A \cap B)$ are disjoint event

$P(A \cup B) = P(A) + P[(\bar{A} \cap B) \cup (A \cap B)] - P(A \cap B)$

$= P(A) + P(B) - P(A \cap B)$

If A and B are mutually exclusive or disjoint event $A \cap B = \emptyset$

$P(A \cup B) = P(A) + P(B)$

If A, B and C are any three events, then

$P(A \cup B \cup C) = P(A)+P(B)+P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)$

If A , B and C are mutually exclusive or disjoint events

$P(A \cup B \cup C) = P(A) + P(B)\ P(C)$

iv) Multiplication theorem of probability:

If A and B are any two event, then

$P(A \cap B) = P(A).\ P(B/A)$

where P(B/A) is conditional probability of occurrence of event B when the event A has already occurred.

or

$P(A \cap B) = P(B).\ P(A/B)$

where P(A/B) is conditional probability of occurrence of event A when the event B has already occur

If event A and B are independent

$P(A \cap B) = P(A).P(B)$

Example 4.1: a) What is the probability that a leap year selected will have 53 Sunday.

b) What is the probability that a non- leap year selected will have 53 Sunday

Solution : a) In leap year total number of days is 366. There will be complete 52 weeks i.e 364 days. 2 days may be

Sunday-Monday, Monday-Tuesday, Tuesday-Wednesday, Wednesday –Thursday, Thursday-Friday, Friday – Saturday, Saturday- Sunday

Exhaustive event (n) = 7

Favourable event (m) = 2 (Sunday-Monday, Saturday- Sunday)

$$p = \frac{2}{7}$$

b) In a non-leap year, total number of days is 365. There will be complete 52 Sunday. One day may be Sunday, Monday, Tuesday, Wednesday, Thursday, Friday, Saturday.

$n = 7$

$m = 1$

$p = \frac{1}{7}$

Example 4.2: Two cards are drawn from a pack of card, what is the probability that they are

i) King

ii) King and queen

iii) Red colour

iv) Heart

Solution: Total cards 52, 2 cards can be drawn in ${}^{52}C_2$ ways

$$n = \frac{52!}{2!(52-2)!} = \frac{52 \times 51 \times 50!}{2 \times 50!} = 26 \times 51$$

i) There are 4 kings and 2 king can be drawn in ${}^{4}C_2$ ways

$$m = {}^{4}C_2 = \frac{4!}{2!(4-2)!} = \frac{4 \times 3 \times 2!}{2 \times 2!} = 6$$

$$p = \frac{6}{26 \times 51} = \frac{1}{13 \times 17} = \frac{1}{221}$$

ii) There are 4 kings and 4 queens. One king and one queen can be drawn in ${}^{4}C_1 \times {}^{4}C_1$ ways

$$m = \frac{4!}{1!(4-1)!} \times \frac{4!}{1!(4-1)!} = \frac{4 \times 3!}{1 \times 3!} \times \frac{4 \times 3!}{1 \times 3!} = 4 \times 4 = 16$$

$$p = \frac{16}{26 \times 51} = \frac{8}{663}$$

iii) There are 26 red cards. Two red colour cards can be drawn in ${}^{26}C_2$ ways

$$m = \frac{26!}{2!(26-2)!} = \frac{26 \times 25 \times 24!}{2 \times 24!} = 13 \times 25$$

$$p = \frac{13 \times 25}{26 \times 51} = \frac{25}{102}$$

iv) There are 13 hearts. Two hearts can be drawn in ${}^{13}C_2$ ways

$$m = \frac{13!}{2!(13-2)!} = \frac{13 \times 12 \times 11!}{2 \times 11!} = 13 \times 6$$

$$p = \frac{13 \times 6}{26 \times 51} = \frac{1}{17}$$

Example 4.3: A card is drawn from a pack of card. What is the probability that it is a

i) King and heart

ii) King or heart

Solution: A card from a pack of card can be drawn in $^{52}C_1$ ways

$$n = \frac{52!}{1!(52-1)!} = \frac{52\times51!}{1\times51!} = 52$$

Let A be the event that selected card is king

$$m = {}^{4}C_1 = \frac{4!}{1!(4-1)!} = 4$$

Let B be the event that selected card is heart

$$m = {}^{13}c_1 = \frac{13!}{1!(13-1)!} = \frac{13!}{1!(13-1)!} = \frac{13\times12!}{1\times12!} = 13$$

$$P(A) = \frac{4}{52}, P(B) = \frac{13}{52}$$

i) King and heart :$P(A\cap B) = P(A).P(B) = \frac{4}{52} \times \frac{13}{52} = \frac{1}{52}$

ii) King or heart $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$$= P(A) + P(B) - P(A)\,P(B)$$

$$= \frac{4}{52} + \frac{13}{52} - \frac{4}{52} \times \frac{13}{52}$$

$$= \frac{4}{52} + \frac{13}{52} - \frac{1}{52} = \frac{16}{52} = \frac{4}{13}$$

Example 4.4 : If $P(A) = \frac{1}{3}$, $P(B) = \frac{3}{4}$, $P(A\cup B) = \frac{11}{12}$, Find P(A/B) and P(B/A)

Solution: $P(A\cup B) = P(A) + P(B) - P(A \cap B)$

$$P(A \cap B) = P(A) + P(B) - P(A\cup B)$$

$$= \frac{1}{3} + \frac{3}{4} - \frac{11}{12}$$

$$= \frac{4+9-11}{12} = \frac{2}{12} = \frac{1}{6}$$

By multiplication theorem

$P(A \cap B) = P(A).\ P(B/A)$

$$P(B/A) = \frac{P(A \cap B)}{P(A)}$$

or

$P(A \cap B) = P(B).\ P(A/B)$

$$P(A/B) = \frac{P(A \cap B)}{P(B)}$$

$$P(B/A) = \frac{\frac{1}{6}}{\frac{1}{3}} = \frac{1}{2}$$

$$P(A/B) = \frac{\frac{1}{6}}{\frac{3}{4}} = \frac{2}{9}$$

Example 4.5: Given P(A) = 0.3, P(B) = 0.78. P(A ∩B) = 0.16

Find i) P(A∪B) ii) P(A ∩ $\overline{B}$) iii) P($\overline{A}$ ∪ $\overline{B}$) iv) P($\overline{A}$ ∩ B) v) P($\overline{A}$ ∩ $\overline{B}$)

i) P(A∪B) = P(A) + P(B) – P(A ∩B) = 0.30 + 0.78 – 0.16 = 0.92

ii) P(A ∩ $\overline{B}$) = P(A) – P(A ∩B) = 0.30 – 0.16 = 0.14

iii) P($\overline{A}$ ∪ $\overline{B}$) = $P\overline{A \cap B)}$ = 1 –P(A ∩B) = 1 – 0.16 = 0.84

iv) P($\overline{A}$ ∩ B) = P(B) – P(A ∩B) = 0.78 – 0.16 = 0.62

v) P($\overline{A}$ ∩ $\overline{B}$) = P($\overline{A \cup B)}$ = 1 – P(A∪B) = 1 – 0.92 = 0.08

Example 4.6: Probability that a horse A will win the race is $\frac{1}{3}$ and that of B is $\frac{1}{6}$. What is the probability that atleast one of them will win the race when they are running in

i) different race

ii) same race.

Solution:

$P(A) = \frac{1}{3}$, $P(B) = \frac{1}{6}$

i) When horses are running in different race:

Probability of atleast one will win the race

P(A∪B) = P(A) + P(B) – P(A ∩B)

$$= \frac{1}{3} + \frac{1}{6} - \frac{1}{3} \times \frac{1}{6} = \frac{6+3-1}{18} = \frac{8}{18} = \frac{4}{9}$$

ii) When horses are running in the same race: A and B will be mutually exclusive i.e

P(A ∩B) = ϕ

$$P(A \cup B) = P(A) + P(B) = \frac{1}{3} + \frac{1}{6} = \frac{2+1}{6} = \frac{1}{2}$$

Example 4.7: From a group of 5 boys and 3 girls, a committee of 3 students is to be constituted. What is the probability that

i) committee consists of no girl

ii) exactly one girl

iii) atleast one girl

iv) more girls than boys

Solution: Number of boys = 5, Number of girls = 3

Total students = 5 + 3 = 8, n = $^8C_3$ = 56

i) No girl i.e. all committee members consist of boys m = $^5c_3$

$p = \frac{^5c_3}{^8c_3} = \frac{10}{56} = \frac{5}{28}$

ii) Committee consists of 2 boys and 1 girl

$p = \frac{^5c_2 \times ^3c_1}{^8c_3} = \frac{10\times3}{56} = \frac{15}{28}$

iii) 1 – no girl selected

$p = 1 - \frac{5}{28} = \frac{23}{28}$

iv) Two girls and one boy or all three girls

$p = \frac{^5c_1 \times ^3c_2}{^8c_3} + \frac{^5c_0\times^3c_3}{^8c_3} = \frac{5\times3}{56} + \frac{1}{56} = \frac{16}{56} = \frac{2}{7}$

Example 4.8: A and B tosses a coin alternately and the person who throw first head will win the game. If A starts the game, find the probability of winning of A and B

Solution:

i) P(H) = $\frac{1}{2}$ P(T) = $\frac{1}{2}$

A start tossing the coin

Since they alternately tosses the coin , A will win if H comes, or if T comes, then for B –T, A–H or A–T,B-T,A-T,B-T, A-H, or so on

H or TTH or TTTTH or TTTTTTH or…

$p = \frac{1}{2} + \frac{1}{2}\times\frac{1}{2}\times\frac{1}{2} + \frac{1}{2}\times\frac{1}{2}\times\frac{1}{2}\times\frac{1}{2}\times\frac{1}{2} + \frac{1}{2}\times\frac{1}{2}\times\frac{1}{2}\times\frac{1}{2}\times\frac{1}{2}\times\frac{1}{2}\times\frac{1}{2} + \ ...$

It is a infinite GP series with first term a = $\frac{1}{2}$ and common ratio r = $\frac{1}{4}$ (r <1)

Sum of an infinite GP series with first term as ‘a’ and common ratio as ‘r’ is

$\text{Sum} = \frac{a}{1-r} = \frac{\frac{1}{2}}{1-\frac{1}{4}} = \frac{\frac{1}{2}}{\frac{3}{4}} = \frac{2}{3}$

$p = \frac{2}{3}$

ii) A-T B-H or A-TB -T A-T B-H or A-TB -T A-T B-T A-T B-H or.....

TH or TTTH or TTTTTH or…

$p = \frac{1}{2}\times\frac{1}{2} + \frac{1}{2}\times\frac{1}{2}\times\frac{1}{2}\times\frac{1}{2} + \frac{1}{2}\times\frac{1}{2}\times\frac{1}{2}\times\frac{1}{2}\times\frac{1}{2}\times\frac{1}{2} + \ldots$

It is an infinite GP series with first term $a = \frac{1}{4}$ and common ratio $r = \frac{1}{4}$ $(r < 1)$

$$\text{Sum} = \frac{a}{1-r} = \frac{\frac{1}{4}}{1-\frac{1}{4}} = \frac{\frac{1}{4}}{\frac{3}{4}} = \frac{1}{3}$$

or

$$P(B) = 1 - P(A) = 1 - \frac{2}{3} = \frac{1}{3}$$

$$p = \frac{1}{3}$$

Example 4.9: Two persons A and B alternately cut a pack of card. A starts the game and is continued until one cuts a king. What is the probability that

i) A will first cut the king

ii) B will first cut the king

Solution: Let E_1 and E_2 be the event that A and B first cut the king

$$P(E_1) = P(E_2) = \frac{1}{13}$$

$$P(\overline{E_1}) = P(\overline{E_2}) = 1 - \frac{1}{13} = \frac{12}{13}$$

If A start the game, he first cut king in following mutually exclusive ways

E_1 or $\overline{E_1} \cap \overline{E_2} \cap E_1$ or $\overline{E_1} \cap \overline{E_2} \cap \overline{E_1} \cap \overline{E_2} \cap E_1$ or $\overline{E_1} \cap \overline{E_2} \cap \overline{E_1} \cap \overline{E_2} \cap \overline{E_1} \cap \overline{E_2} \cap E_1$ ….

$P(E_1) + P(\overline{E_1} \cap \overline{E_2} \cap E_1) + P(\overline{E_1} \cap \overline{E_2} \cap \overline{E_1} \cap \overline{E_2} \cap E_1) + P(\overline{E_1} \cap \overline{E_2} \cap \overline{E_1} \cap \overline{E_2} \cap \overline{E_1} \cap \overline{E_2} \cap E_1) + \ldots$

$$= \frac{1}{13} + \frac{12}{13}\times\frac{12}{13}\times\frac{1}{13} + \frac{12}{13}\times\frac{12}{13}\times\frac{12}{13}\times\frac{12}{13}\times\frac{1}{13} + \frac{12}{13}\times\frac{12}{13}\times\frac{12}{13}\times\frac{12}{13}\times\frac{12}{13}\times\frac{1}{13} + \ldots$$

It is a GP series with $a = \frac{1}{13}$, $r = \frac{12}{13}\times\frac{12}{13} = \frac{144}{169}$

$$\text{Sum} = \frac{\frac{1}{13}}{1-\frac{144}{169}} = \frac{\frac{1}{13}}{\frac{25}{169}} = \frac{13}{25}$$

Probability that first B cut king $= 1 - \frac{13}{25} = \frac{12}{25}$

Example 4.10: Probability that a 50 year old man will be alive at the age of 60 is 0.80 and probability that 45 year old women will be alive at the age of 55 is 0.90. A man whose age is 50 and his wife is 45, what is the probability that

i) both will be alive after 10 years

ii) none will be alive after 10 year

iii) exactly one will be alive after 10 years

iv) atleast one will be alive after 10 years

v) atmost one will be alive after 10 years

Solution:

Let the man is A and women is B then $P(A) = 0.8, P(B) = 0.9$

$P(\overline{A}) = 1 - P(A) = 1 - 0.8 = 0.2$

$P(\overline{B}) = 1 - P(B) = 1 - 0.9 = 0.1$

i) Probability that both will be alive after 10 years

$P(A \cap B) = P(A).P(B) = 0.8 \times 0.9 = 0.72$

ii) None will be alive after 10 years $= P(\overline{A} \cap \overline{B})$

$P(\overline{A} \cap \overline{B}) = = P(\overline{A}) . P(\overline{B}) = 0.2 \times 0.1 = 0.02$

iii) Exactly one will be alive after 10 years

A alive and B die or A die and B alive

$P(A \cap \overline{B}) + P(\overline{A} \cap B)$

$P(A). P(\overline{B}) + P(\overline{A}). P(B) = 0.8 \times 0.1 + 0.2 \times 0.9 = 0.08 + 0.18 = 0.26$

iv) Atleast one will be alive = 1 – none will be alive

$= 1 - 0.02 = 0.98$

or

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$= 0.8 + 0.9 - 0.8 \times 0.9 = 0.98$

or

Exactly one alive + both alive = 0.26 + 0.72 = 0.98

v) Atmost 1 alive = none alive + exactly one alive = 0.02 + 0.26 = 0.28

or $1 - P(\text{both alive}) = 1 - P(A \cap B) = 1 - 0.72 = 0.28$

Example 4.11: Three papers A,B and C are published from a city. It is established from the survey that 20% read A, 16 % read B, 14 % read C, 8% read A and B, 5 % read A and C, 4 % read B and C, 2% read all three news paper. A person is selected randomly what is the probability that

i) does not read any paper

ii) does not read C

iii) read A but not B

Solution:

Given that P(A) = 0.20, P(B) = 0.16, P(C) = 0.14, P(A ∩B) = 0.08, P(A ∩C) = 0.05, P(B ∩C) = 0.04, P(A ∩B∩ C) = 0.02

i) Does not read any paper $P(\bar{A} \cap \bar{B} \cap \bar{C}) = P(\overline{A \cup B \cup C})$

P(A∪B∪ C) = P(A) +P(B) +P(C) – P(A ∩B) – P(A ∩C) – P(B ∩C)+ P(A ∩B∩ C)

= 0.20 + 0.16+ 0.14 – 0.08 – 0.05 – 0.04 + 0.02 = 0.35

$P(\overline{A \cup B \cup C}) = 1 - P(A \cup B \cup C) = 1 - 0.35 = 0.65$

Probability that a person will not read any paper = 0.65

ii) Does not read C = $P(\bar{C}) = 1 - P(C) = 1 - 0.14 = 0.86$

iii) Read A but not B

$P(A \cap \bar{B}) = P(A) - P(A \cap B) = 0.20 - 0.08 = 0.12$

Example 4.12: If $P(A \cup B) = \frac{5}{6}$, $P(A \cap B) = \frac{1}{3}$, $P(\bar{B}) = \frac{1}{2}$. find P(A) and P(B). show that A and B are independent

Solution:

$P(B) = 1 - P(\bar{B}) = 1 - \frac{1}{2} = \frac{1}{2}$

P(A∪B) = P(A) + P(B) – P(A ∩B)

P(A) = P(A∪B) – P(B) + P(A ∩B)

$= \frac{5}{6} - \frac{1}{2} + \frac{1}{3} = \frac{5-3+2}{6} = \frac{4}{6} = \frac{2}{3}$

A and B are independent if $P(A \cap B) = P(A).P(B) = \frac{1}{2} \times \frac{2}{3} = \frac{1}{3} = P(A \cap B)$

Since P(A ∩B) = P(A).P(B), hence A and B are independent.

Example 4.13: A bag contains 7 pear, 6 peach and 5 chikoo. Two fruits are selected. What is the probability that both of them are pear

Solution: Total number of fruits = 7 + 6 + 5 = 18

$n = {}^{18}C_2 = \frac{18!}{2!(18-2)!} = \frac{18 \times 17 \times 16!}{2!(16)!} = 9 \times 17 = 153$

Total pear = 7

$m = {}^{7}C_2 = \frac{7!}{2!(7-2)!} = \frac{7 \times 6 \times 5!}{2!(5)!} = 7 \times 3 = 21$

$p = \frac{21}{153} = \frac{7}{51}$

Example 4.14: A bag contains 10 apple fruits of which 2 are red, 3 are green and rest are yellowish. Three fruits are selected what is the probability that

i) they are of different colour

ii) 2 fruits are of same colour

iii) all fruits are of same colour

Solution: Total fruits 10, 3 fruits can be selected in ${}^{10}C_3$ ways $= \frac{10!}{3!\times 7!} = \frac{10\times 9\times 8\times 7!}{3\times 2\times 1\times 7!} = 120$

$n = 120$

Red = 2, Green = 3, Yellow = $10 - 2 - 3 = 5$

i) All fruits are of different colour i.e 1 red and I green and 1 yellow fruit

$m = {}^2C_1 \times {}^3C_1 \times {}^5C_1 = 2 \times 3 \times 5 = 30$

$p = \frac{30}{120} = \frac{1}{4}$

ii) Two fruits are of same colour

2 red and 1 from remaining fruits (8) or 2 green and 1 from remaining fruits (7) or 2 yellow and 1 from remaining fruits (5)

Favourable cases : 2 Red and 1 other $= {}^2C_2 \times {}^8C_1 = 1 \times 8 = 8$

2 green and 1 other $= {}^3C_2 \times {}^7C_1 = 3 \times 7 = 21$

2 Yellow and 1 other $= {}^5C_2 \times {}^5C_1 = 10 \times 5 = 50$

$p = \frac{8}{120} + \frac{21}{120} + \frac{50}{120} = \frac{79}{120}$

iii) All are of same colour

3 may be green or three may be yellow

Favourable cases: Green ${}^3C_3 = 1$, Yellow ${}^5C_3 = 10$

$p = \frac{1}{120} + \frac{10}{120} = \frac{11}{120}$

Example 4.15: A bag contains 12 fruits of which 2 fruits are spoiled. 4 fruits are selected with replacement. What is the probability that

i) no spoiled fruit

ii) exactly one spoiled fruit

Solution: Total fruits 12, spoiled fruit =2

Probability that a selected fruit is spoiled $p = \frac{2}{12} = \frac{1}{6}$

Probability that selected fruit is not spoiled q = $1-\frac{1}{6}=\frac{5}{6}$

i) Probability of no spoiled fruit

$$P(X=0) = {}^{4}C_0 \left(\frac{1}{6}\right)^0 \left(\frac{5}{6}\right)^4 = 1 \times 1 \times \frac{625}{1296} = \frac{625}{1296}$$

ii) Probability of 1 spoiled fruit

$$P(X=1) = {}^{4}C_1 \left(\frac{1}{6}\right)^1 \left(\frac{5}{6}\right)^{4-1} = 4 \times \frac{125}{1296} = \frac{500}{1296}$$

Example 4.16: A bag contains 6 red and 9 green apples. Two fruits are drawn at a time. What is the probability that first draw gives two red and second draw gives 2 green apple when

i) Fruits are not replaced back before the second draw

ii) Fruit are replaced back before the second draw.

Solution: Let A be the event that first draw gives 2 red fruit, B be the event that second draw gives 2 green fruits

i) $P(A) = \frac{{}^{6}c_2}{{}^{15}c_2} = \frac{15}{105} = \frac{1}{7}$

When fruits are not replaced back before the second draw, remaining fruits are 13

$$P(B/A) = \frac{{}^{9}c_2}{{}^{13}c_2} = \frac{36}{78} = \frac{6}{13}$$

$$P(A \cap B) = P(A).P(B/A) = \frac{1}{7} \times \frac{6}{13} = \frac{6}{91}$$

ii) When fruits are replaced back before the second draw

$$P(B/A) = \frac{{}^{9}c_2}{{}^{15}c_2} = \frac{36}{105} = \frac{12}{35}$$

$$P(A \cap B) = P(A).P(B) = \frac{1}{7} \times \frac{12}{35} = \frac{12}{245}$$

Example 4.17: A bag contains 6 red and 9 green apple. Two fruits are drawn at a time. What is the probability that first draw gives two red and second draw gives 2 red apple when

i) Fruits are not replaced back before the second draw

ii) Fruit are replaced back before the second draw.

Solution: Let A be the event that first draw gives 2 red fruit, B be the event that second draw gives 2 red fruits

i) $P(A) = \frac{{}^{6}c_2}{{}^{15}c_2} = \frac{15}{105} = \frac{1}{7}$

When fruits are not replaced back before the second draw, remaining fruits are 13 and red fruits are 4

$$P(B/A) = \frac{^4c_2}{^{13}c_2} = \frac{6}{78} = \frac{1}{13}$$

$$P(A \cap B) = P(A).P(B/A) = \frac{1}{7} \times \frac{1}{13} = \frac{1}{91}$$

ii) When fruits are replaced back before the second draw

$$P(B/A) = P(B) = \frac{^6c_2}{^{15}c_2} = \frac{15}{105} = \frac{1}{7}$$

$$P(A \cap B) = P(A).P(B) = \frac{1}{7} \times \frac{1}{7} = \frac{1}{49}$$

Example 4.18: A box contains 12 fruits of which 3 are spoiled. Fruits are selected one by one and not put back. What is the probability that seventh fruit selected is the last spoiled fruit.

Solution: Total fruits = 12 Spoiled fruit = 3 Good fruit = 12 –3 = 9

Let A be the event that 2 spoiled fruits are selected in first six draw

B be the event that seventh fruit selected is spoiled fruit

$$P(A) = \frac{^9c_4 \times ^3c_2}{^{15}c_6} = \frac{126 \times 3}{5005} = \frac{378}{5005}$$

Now 6 fruits have been selected, remaining fruits = 12 – 6 = 6

$$P(B) = \frac{1}{6}$$

$$P(A \cap B) = \frac{378}{5005} \times \frac{1}{6} = \frac{63}{5005}$$

Example 4.19: Probability that a person A will solve the problem is $\frac{1}{5}$ and that of B is $\frac{1}{4}$, both are solving independently. What is the probability that

i) Both will solve the problem

ii) None will solve the problem

iii) Exactly one will solve the problem

iv) Atmost one will solve the problem

v) Atleast one will solve the problem

Solution:

$$P(A) = \frac{1}{5}, P(\bar{A}) = 1 - \frac{1}{5} = \frac{4}{5}$$

$$P(B) = \frac{1}{4}, P(\bar{B}) = 1 - \frac{1}{4} = \frac{3}{4}$$

i) **Both will solve the problem**: A is solving and B is solving the problem i.e. $A \cap B$

$P(A \cap B) = P(A).P(B) = \frac{1}{5} \times \frac{1}{4} = \frac{1}{20}$

ii) **None will solve the problem:** A is not solving and B is not solving i.e. $\bar{A} \cap \bar{B}$

$P(\bar{A} \cap \bar{B}) = P(\bar{A}) \, . \, P(\bar{B}) = \frac{4}{5} . \frac{3}{4} = \frac{3}{5}$

iii) **Exactly one will solve the problem:** A is solving and B is not solving or A is not solving and B is solving the problem i.e. $(A \cap \bar{B}) \cup (\bar{A} \cap B)$

$P((A \cap \bar{B}) \cup (\bar{A} \cap B)) = P(A). P(\bar{B}) + P(\bar{A}) \, . P(B)$

$= \frac{1}{5} \times \frac{3}{4} + \frac{4}{5} \times \frac{1}{4} = \frac{3+4}{20} = \frac{7}{20}$

iv) **Atmost one will solve the problem:** Either none will solve or exactly one will solve the problem

$P(\text{Atmost one will solve the problem}) = P(\bar{A} \cap \bar{B}) + P((A \cap \bar{B}) \cup (\bar{A} \cap B))$

$= \frac{3}{5} + \frac{7}{20} = \frac{12+7}{20} = \frac{19}{20}$

or

= 1 – both will solve the problem

$= 1 - \frac{1}{20} = \frac{19}{20}$

v) Atleast one will solve the problem

$P(A \cup B) = P(A) + (B) - P(A \cap B)$

$= \frac{1}{5} + \frac{1}{4} - \frac{1}{5} \times \frac{1}{4} = \frac{4+5-1}{20} = \frac{8}{20} = \frac{2}{5}$

or

= 1 – none will solve the problem

$= 1 - P(\bar{A} \cap \bar{B})$

$= 1 - \frac{3}{5} = \frac{2}{5}$

or

Exactly one will solve the problem or both will solve the problem

$P((A \cap \bar{B}) \cup (\bar{A} \cap B)) + P(A \cap B)$

$= \frac{7}{20} + \frac{1}{20} = \frac{7+1}{20} = \frac{8}{20} = \frac{2}{5}$

Example 4.20: Let A and B are any two event $P(A \cup B) = 0.7$, $P(A) = 0.4$, $P(B) = p$. For what choice of p

i) Event A and B are mutually exclusive

ii) Event A and B are independent

Solution:

i) If Event A and B are mutually exclusive

$P(A \cup B) = P(A) + P(B)$

$0.7 = 0.4 + p$

$p = 0.3$

If p =0.3, event A and B will be mutually exclusive

ii) If A and B are independent then $P(A \cap B) = P(A).P(B)$

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$= P(A) + P(B) - P(A).P(B)$

$0.7 = 0.4 + p - 0.4p$

$0.3 = 0.6p$

$p = 0.5$

If p =0.5, event A and B will be independent.

Example 4.21: A person A speak truth 3 times out of 5 times, while B speak truth 2 times out of 7 times. What is the probability that they will contradict each other for stating the same fact.

Solution:

P(A speak truth) $= \frac{3}{5}$ $P(\overline{A}) = 1 - \frac{3}{5} = \frac{2}{5}$

P(B speak truth) $= \frac{2}{7}$ $P(\overline{B}) = 1 - \frac{2}{7} = \frac{5}{7}$

They will contradict each other if A speak truth and B speak lie or A speak lie and B speak truth

$P((A \cap \overline{B}) \cup (\overline{A} \cap B)) = \frac{3}{5} \times \frac{5}{7} + \frac{2}{5} \times \frac{2}{7} = \frac{15+4}{35} = \frac{19}{35}$

Example 4.22: A problem is given to three students A, B and C, they are solving independently whose chances of solving the problem are $\frac{1}{4}$, $\frac{1}{5}$ and $\frac{1}{3}$ respectively. What is the probability that

i) None will solve the problem

ii) All will solve the problem

iii) Exactly one will solve the problem

iv) Atmost one will solve the problem

v) Exactly two will solve the problem

vi) Atleast one will solve the problem

vii) Atmost two will solve the problem

Solution:

$P(A) = \frac{1}{4}$, $P(\bar{A}) = \frac{3}{4}$

$P(B) = \frac{1}{5}$, $P(\bar{B}) = \frac{4}{5}$

$P(C) = \frac{1}{3}$, $P(\bar{C}) = \frac{2}{3}$

i) None will solve the problem

$P(\bar{A} \cap \bar{B} \cap \bar{C}) = P(\bar{A})\, P(\bar{B})\, P(\bar{C})$

$P(\bar{A} \cap \bar{B} \cap \bar{C}) = \frac{3}{4} \times \frac{4}{5} \times \frac{2}{3} = \frac{2}{5}$

ii) All will solve the problem

$P(A \cap B \cap C) = P(A)\, P(B)\, P(C)$

$= \frac{1}{4} \times \frac{1}{5} \times \frac{1}{3} = \frac{1}{60}$

iii) Exactly one will solve the problem

$P(A \cap \bar{B} \cap \bar{C}) + P(\bar{A} \cap B \cap \bar{C}) + P(\bar{A} \cap \bar{B} \cap C) = P(A)P(\bar{B})P(\bar{C}) + P(\bar{A})P(B)P(\bar{C}) + P(\bar{A})\, P(\bar{B})\, P(C)$

$= \frac{1}{4} \times \frac{4}{5} \times \frac{2}{3} + \frac{3}{4} \times \frac{1}{5} \times \frac{2}{3} + \frac{3}{4} \times \frac{4}{5} \times \frac{1}{3}$

$= \frac{8+6+12}{60} = \frac{26}{60} = \frac{13}{30}$

iv) Atmost one will solve the problem = none will solve the problem + exactly one will solve the problem

P(Atmost one will solve the problem) = $\frac{2}{5} + \frac{13}{30} = \frac{12+13}{30} = \frac{25}{30} = \frac{5}{6}$

v) Exactly two will solve the problem =

$P(A \cap B \cap \bar{C}) + P(\bar{A} \cap B \cap C) + P(A \cap \bar{B} \cap C) = P(A)P(B)P(\bar{C}) + P(\bar{A})P(B)P(C) + P(A)\, P(\bar{B})\, P(C)$

$= \frac{1}{4} \times \frac{1}{5} \times \frac{2}{3} + \frac{3}{4} \times \frac{1}{5} \times \frac{1}{3} + \frac{1}{4} \times \frac{4}{5} \times \frac{1}{3} = \frac{2+3+4}{60} = \frac{9}{60} = \frac{3}{20}$

vi) Atleast one will solve the problem = exactly one will solve the problem + exactly two will solve the problem + all will solve the problem

P(Atleast one will solve the problem) = $\frac{13}{30} + \frac{3}{20} + \frac{1}{60} = \frac{26+9+1}{60} = \frac{26+9+1}{60} = \frac{36}{60} = \frac{3}{5}$

or

1 – P(none will solve the problem) = $1 - \frac{2}{5} = \frac{3}{5}$

or

$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)$

$= \frac{1}{4} + \frac{1}{5} + \frac{1}{3} - \frac{1}{4} \times \frac{1}{5} - \frac{1}{4} \times \frac{1}{3} - \frac{1}{5} \times \frac{1}{3} + \frac{1}{4} \times \frac{1}{5} \times \frac{1}{3}$

$= \frac{15+12+20-3-5-4+1}{60} = \frac{36}{60} = \frac{3}{5}$

vii) Atmost two will solve the problem = none will solve the problem + exactly one will solve the problem + exactly two will solve the problem

P(Atmost two will solve the problem)$= \frac{2}{5} + \frac{13}{30} + \frac{3}{20} = \frac{24+26+9}{60} = \frac{59}{60}$

or

1 – P(all will solve the problem) $= 1 - \frac{1}{60} = \frac{59}{60}$

Example 4.23 : A box contains 10 good fruits and 6 damaged fruit. Two successive drawn of 2 fruits each are made such that

i) Fruits are not replaced back before the second draw

ii) Fruits are replaced back before the second draw

Find the probability that first draw gives 2 damaged fruit and second draw gives 2 damaged fruit

Solution: Let A is the event that first draw is giving 2 damaged fruit

B is the event that second draw gives two damaged fruit

Total fruits = 16

i) $P(A) = \frac{{}^{6}c_2}{{}^{16}c_2} = \frac{15}{120} = \frac{1}{8}$

When fruits are not replaced back before the second draw, damaged fruit is 4 and total fruit is 14

$P(B/A) = \frac{{}^{4}c_2}{{}^{14}c_2} = \frac{6}{91}$

$P(A \cap B) = P(A).P(B/A)$

$= \frac{1}{8} \times \frac{6}{91} = \frac{3}{364}$

ii) When fruits are replaced back before the second draw, and total fruit is 16

$P(B/A) = P(B) = \frac{{}^{6}c_2}{{}^{16}c_2} = \frac{15}{120} = \frac{1}{8}$

$P(A \cap B) = P(A).P(B)$

$$= \frac{1}{8} \times \frac{1}{8} = \frac{1}{64}$$

Example 4. 24: A box contains 10 good fruits and 6 damaged fruit. Two successive drawn of 2 fruits each are made such that

i) Fruits are not replaced back before the second draw

ii) Fruits are replaced back before the second draw

Find the probability that first draw gives 2 damaged fruit and second draw gives 2 good fruit

Solution:

Let A is the event that first draw is giving 2 damaged fruit

B is the event that second draw gives two good fruit

Total fruits = 16

$$P(A) = \frac{^6c_2}{^{16}c_2} = \frac{15}{120} = \frac{1}{8}$$

i) When fruits are not replaced back before the second draw, total fruit is 14

$$P(B/A) = \frac{^{10}c_2}{^{14}c_2} = \frac{45}{91}$$

$$P(A \cap B) = P(A).P(B/A)$$

$$= \frac{1}{8} \times \frac{45}{91} = \frac{45}{724}$$

ii) When fruits are replaced back before the second draw, total fruit is 16

$$P(B/A) = P(B) = \frac{^{10}c_2}{^{16}c_2} = \frac{45}{120} = \frac{3}{8}$$

$$P(A \cap B) = P(A).P(B)$$

$$= \frac{1}{8} \times \frac{3}{8} = \frac{3}{64}$$

Example 4.25: A box contains 15 fruits of which 3 are damaged fruits. The fruits are selected one by one and examined. The examined fruit are not replaced back before the next draw. Find the probability that the fifth one examined is the last damaged fruits.

Solution:

There are 3 damaged and 12 good fruits in the box.

Let A is the event of getting exactly 2 damaged fruits in first 4 draw

B is the event that fifth examined fruit is last defective.

$$P(A) = \frac{^3c_2 \times ^{12}c_2}{^{15}c_4} = \frac{3 \times 66}{1365} = \frac{198}{1365}$$

Total fruits left = 11

Damaged fruit = 1

$$P(B/A) = \frac{1}{11}$$

$$P(A \cap B) = P(A).P(B/A) = \frac{198}{1365} \times \frac{1}{11} = \frac{6}{455}$$

Example 4.26: A box contain 10 good and 6 damaged fruits. A person draw three fruits from the box at random. What is the probability that among the fruits drawn there is atleast one fruit of each type

Solution: 3 fruits can be drawn as one good and 2 damaged fruit or two good and one damaged fruit

$$p = \frac{{}^{10}c_1 \times {}^{6}c_2}{{}^{16}c_3} + \frac{{}^{10}c_2 \times {}^{6}c_1}{{}^{16}c_3}$$

$$= \frac{10 \times 15}{560} + \frac{45 \times 6}{560} = \frac{150+270}{560} = \frac{420}{560} = \frac{3}{4}$$

Example 4.27: Following is the distribution of livestock in 35 families. One family is selected Find the probability that

i) family has 4 livestock

ii) atmost 3 livestock

iii) atleast 5 livestock

Number of cattle (X)	0	1	2	3	4	5	6	7	8
Frequency (f)	2	1	4	5	5	5	4	6	3

Solution:

i) There are 5 families which have 4 livestock, any of these family can be selected
P(family has 4 livestock) = $\frac{5}{35} = \frac{1}{7}$

ii) P(Atmost 3 livestock) = no or 1 or 2 or 3 livestocks i.e. = $\frac{2}{35} + \frac{1}{35} + \frac{4}{35} + \frac{5}{35} = \frac{12}{35}$

iii) P(Atleast five livestock) = 5 or 6 or 7 livestocks i.e. = $\frac{5}{35} + \frac{4}{35} + \frac{6}{35} + \frac{3}{35} = \frac{18}{35}$

Example 4. 28: If $P(A \cap B) = \frac{1}{2}$, $P(\overline{A} \cap \overline{B}) = \frac{2}{3}$, P(A) = p, P(B) = 2p, find value of p

Solution: $P(\overline{A \cup B}) = P(\overline{A} \cap \overline{B})$

$$P(\overline{A \cup B}) = \frac{2}{3}$$

$P(A \cup B) = 1 - P(\overline{A \cup B}) = 1 - \frac{2}{3} = \frac{1}{3}$

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$\frac{1}{3} = p + 2p - \frac{1}{2}$

$3p = \frac{1}{3} + \frac{1}{2} = \frac{2+3}{6} = \frac{5}{6}$

$p = \frac{5}{18}$

4.5 Random Variable: Suppose S is the sample space of a given random experiment. A real valued function define on the sample space and taking value from $-\infty$ to $+\infty$ is known as random variable. Random variable is denoted by capital letter X,Y,Z,… etc . Spectrum of a random variable means all possible values of the random variable. Random variable is of two types.

i) **Discrete Random Variable:** A random variable which takes countable finite number of values or which is defined on a discrete sample space.

Example: A coin is tossed three times. Let X denotes number of times head is occurring. Sample space consists of 8 sample points as

Sample point:	HHH	HHT	HTH	HTT	THH	THT	TTH	TTT
X :	3	2	2	1	2	1	1	0

Random Variable X will take value 0, 1, 2 or 3.

X : 0 1 2 3

ii) **Continuous Random Variable:** A random variable which takes all possible values in a given range is known as continuous random variable.

4.5.1 Probability mass function: Suppose X is a discrete random variable taking countably infinite number of values x_1, x_2, x_3, ... With each possible outcome x_i, we associate a number $p_i = P(X = x_i)$, such that it must satisfy the following conditions

i) $Pi \geq 0$

ii) $\sum p_i = 1$

Then p_i is known as probability mass function of random variable X

- **Distribution function:** It indicates the probability of a random variable from $-\infty$ to a value of that random variable.It is denoted by $F(x) = P(X \leq x)$

Suppose a coin is tossed three times. Let X denotes number of times head is occurring. Then probability distribution of X is

X :	**0**	**1**	**2**	**3**

P(X = x):	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$
F(x) :	$\frac{1}{8}$	$\frac{4}{8}$	$\frac{7}{8}$	1

Example 4.29 : Suppose X is a random variable taking values

X	-2	-1	0	1	2
P(X =x)	K	2k	3k	3k	k

i) Find value of k such that it is a probability mass function

ii) $P(-1 < X \leq 1)$

iii) $P(-1 \leq X \leq 1)$

iv) $P(-1 < X < 1)$

Solution:

i) If X is a random variable, the sum of all probabilitities must be equal to 1 i.e.

$k + 2k + 3k + 3k + k = 1$

$10\,k = 1$

$k = \frac{1}{10}$

X	-2	-1	0	1	2
P(X =x)	$\frac{1}{10}$	$\frac{2}{10}$	$\frac{3}{10}$	$\frac{3}{10}$	$\frac{1}{10}$

i) $P(-1 < X \leq 1) = P(X = 0) + P(X = 1)$

$$= \frac{3}{10} + \frac{3}{10} = \frac{6}{10}$$

ii) $P(-1 \leq X \leq 1) = P(X = -1) + P(X = 0) + P(X = 1)$

$$= \frac{2}{10} + \frac{3}{10} + \frac{3}{10} = \frac{8}{10}$$

iii) $P(-1 < X < 1) = P(X = 0) = \frac{3}{10}$

Example 4.30: Random variable X has following probability distribution

X	-3	-2	-1	0	1	2	3
P(X = x)	K	2k	2k	3k	k^2	$2k^2$	$7k^2+k$

i) Find value of k

ii) $P(X<0)$

iii) $P(-2 < X \leq 1)$

iv) $P(-1 \leq X \leq 2)$

v) $P(-1 < X < 2)$

Solution: i) If X is a random variable, the sum of all probabilitities must be equal to 1

i.e. $k + 2k + 2k + 3k + k^2 + 2k^2 + 7k^2 + k = 1$

$10k^2 + 9k = 1$

$10k^2 + 9k - 1 = 0$

$10k^2 + 10k - k - 1 = 0$

$10k(k + 1) - 1(k + 1) = 0$

$(k + 1)(10k - 1) = 0$

$k + 1 = 0 \Rightarrow k = -1$ or $10k - 1 = 0 \Rightarrow k = \frac{1}{10}$

$k = -1$ rejected as probability cannot be negative, hence $k = \frac{1}{10}$

X	-3	-2	-1	0	1	2	3
P(X = x)	$\frac{1}{10}$	$\frac{2}{10}$	$\frac{2}{10}$	$\frac{3}{10}$	$\frac{1}{100}$	$\frac{2}{100}$	$\frac{17}{100}$

ii) $P(X<0) = P(X=-3) + P(X=-2) + P(X=-1) = \frac{1}{10} + \frac{2}{10} + \frac{2}{10} = \frac{5}{10}$

iii) $P(-2 < X \leq 1) = P(X=-1) + P(X=0) + P(X=1) = \frac{2}{10} + \frac{3}{10} + \frac{1}{100} = \frac{51}{100}$

iv) $P(-1 \leq X \leq 2) = P(X=-1) + P(X=0) + P(X=1) + P(X=2)$

$= \frac{2}{10} + \frac{3}{10} + \frac{1}{100} + \frac{2}{100} = \frac{53}{100}$

v) $P(-1 < X < 2) = P(X=0) + P(X=1) = \frac{3}{10} + \frac{1}{100} = \frac{31}{100}$

Example 4.31: A box contains 10 fruits of which 3 are spoiled. A sample of 4 fruits is drawn from box at random without replacement. Let X denotes number of spoiled fruits. Find probability distribution of X.

Solution: Total Fruits = 10

Spoiled fruit =3

Good fruits $= 10 - 3 = 7$

Now 4 fruits will be selected from the box, spoiled fruits may be 0,1,2,3

$$P(X=0) = \frac{{}^{7}c_4}{{}^{10}c_4} = \frac{\frac{7!}{4!\times 3!}}{\frac{10!}{4!\times 6!}} = \frac{\frac{7\times 6\times 5\times 4!}{3\times 2\times 1\times 4!}}{\frac{10\times 9\times 8\times 7\times 6!}{4\times 3\times 2\times 1\times 6!}} = \frac{35}{210} = \frac{1}{6}$$

$$P(X=1) = \frac{{}^{7}c_3 \times {}^{3}c_1}{{}^{10}c_4} = \frac{\frac{7!}{4!\times 3!}\times 3}{\frac{10!}{4!\times 6!}} = \frac{\frac{7\times 6\times 5\times 4!}{3\times 2\times 1\times 4!}\times 3}{\frac{10\times 9\times 8\times 7\times 6!}{4\times 3\times 2\times 1\times 6!}} = \frac{35\times 3}{210} = \frac{1}{2}$$

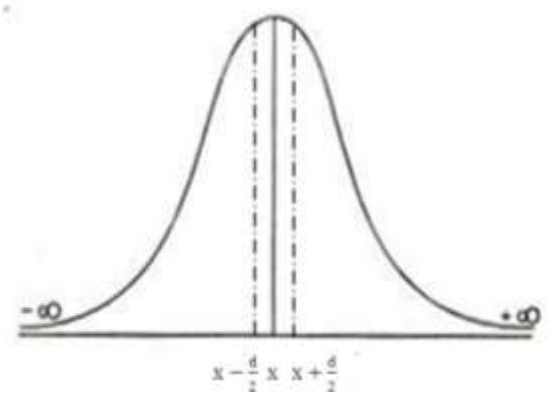
$$P(X=2) = \frac{{}^{7}c_2 \times {}^{3}c_2}{{}^{10}c_4} = \frac{21\times 3}{210} = \frac{3}{10}$$

$$P(X=3) = \frac{{}^{7}c_1 \times {}^{3}c_3}{{}^{10}c_4} = \frac{7\times 1}{210} = \frac{1}{30}$$

X	0	1	1	3
P(X = x)	$\frac{1}{6}$	$\frac{1}{2}$	$\frac{3}{10}$	$\frac{1}{30}$
F (X)	$\frac{1}{6}$	$\frac{2}{3}$	$\frac{29}{30}$	1

1.5.2 Probability density function (pdf): Let X be a continuous random variable. Take small interval of length dx around the point x and f(x) be any continuous function of x such that $P(x - \frac{dx}{2} \leq X \leq x + \frac{dx}{2}) = f(x)dx$, f(x) is known as probability density function of random variable such that

i) $f(x) > 0$ ii) $\int_{-\infty}^{\infty} f(x)dx = 1$ $-\infty < X < \infty$



Example 4.32: Diameter of fruit is given by

$6x(1-x)$ $0 < X < 1$

i) Check that it is a probability density function
ii) Find probability that diameter of fruit is < 0.5
iii) Find probability that diameter of fruit lies between 0.25 to 0.5

Solution: i) $\int_0^1 6x(1-x)\,dx = 1$

$= \int_0^1 6x\,dx - \int_0^1 6x^2\,dx$

$= 6\left[\frac{x^2}{2}\right]_0^1 - 6\left[\frac{x^3}{3}\right]_0^1$

$= 3(1^2 - 0^2) - 2(1^3 - 0^3)$

$= 3 - 2$

$= 1$

Hence it is a pdf.

ii) $P(X < 0.5)$

$\int_0^{0.5} 6x(1-x)\,dx$

$\int_0^{0.5} 6x\,dx - \int_0^{0.5} 6x^2\,dx$

$6\left[\frac{x^2}{2}\right]_0^{0.5} - 6\left[\frac{x^3}{3}\right]_0^{0.5}$

$3(0.5^2 - 0^2) - 2(0.5^3 - 0^3)$

$3\times 0.25 - 2\times 0.125$

$0.75 - 0.25 = 0.5$

iii) $P(0.25 < X < 0.5)$

$3(0.5^2 - 0.25^2) - 2(0.5^3 - 0.25^3)$

$3(0.25 - 0.0625) - 2(0.125 - 0.015625)$

$3 \times 0.1875 - 2 \times 0.10937$

$0.5625 - 0.21875$

0.34375

Probability Distribution Chapter V

5.1 Binomial Distribution: Suppose n independent trials are performed repeatedly and each trial results into two mutually exclusive events, occurrence of an event is known as success and non-occurrence of an event is known as failure. Probability of success remains constant from trial to trial. If there are x successes, then there will be n-x failure, these success and failure may occured in any order

SFFFSSFFSSFSFSS…

- Let probability of success is p and probability of failure is q, then probability of x successes in n independent trial is

  pqqqppqqppqpqpp…

  $= p^x.q^{n-x}$

  But x successes in n trials occurs in ${}^nc_x$ ways

  Hence probability of x successes in n independent trial is given by ${}^nc_x.\ p^x.q^{n-x}$

- A random variable X is having binomial distribution with parameters n and p and its probability mass function (p.m.f.) is given by

  $P(X = x) = {}^nc_x.\ p^x.q^{n-x}$

  $x = 0,1, 2, 3,\ldots, n$

  $0 \le p \le 1$

  $p + q = 1$

  Also $\sum {}^nc_x.\ p^x.q^{n-x} = (q + p)^n = 1$

5.1.1 Physical conditions of binomial distribution:

i) Number of trials (n) are finite

ii) Each trial is independent to each other

iii) Each trial results in two mutually exclusive events i.e. success or failure

iv) Probability of success remains constant from trial to trial

v) It is generally applied when sampling is done with replacement from infinite population so that probability of success remains same.

Example 5.1: A box contains 100 fruits of which 30 fruits are damaged. Six fruits are selected randomly, what is the probability that it contains

i) Exactly 2 damaged fruits

ii) No damaged fruit

iii) Atleast 1 damaged fruit

iv) Atmost 5 damaged fruits

Solution: Probability of damaged fruit = $p = \frac{30}{100} = 0.3$, $q = 1 - 0.3 = 0.7$

i) Exactly 2 fruits are damaged

$P(X = 2) = {}^6C_2 \times 0.3^2 \times 0.7^{(6-2)}$

$= 15 \times 0.09 \times 0.2401 = 0.324$

ii) No damaged fruit

$P(X = 0) = {}^6C_0 \times 0.3^0 \times 0.7^6$

$= 0.7^6 = 0.117$

iii) Atleast one damaged fruit

$= 1 - P(X = 0) = 1 - 0.117649 = 0.882$

iv) Atmost five damaged fruits

$= 1 - P(X = 6)$

$= 1 - {}^6C_6 \times 0.3^6 \times 0.7^{6-6}$

$= 1 - 0.3^6 = 1 - 0.000729 = 0.9993$

5.1.2 Constants of binomial distribution:

Mean	=	np
Variance (μ_2)	=	npq

Note : Mean of binomial distribution is always greater than variance

- **Mode :** There are two cases

i) If (n+1).p is not an integer, distribution is unimodal and integer part of (n+1).p will be the value of mode

ii) If (n+1).p is an integer, distribution is bi-modal, let (n+1).p = m, two mode will be m − 1 and m.

Example 5.2: Mean of binomial distribution is 4 and variance is 3. Find $P(X \geq 2)$ and mode of the distribution

Solution: Mean = np = 4 (i)

Variance = npq = 3 (ii)

Divide (ii) by (i)

$\frac{npq}{np} = \frac{3}{4}$

$q = \frac{3}{4}$, $p = 1 - \frac{3}{4} = \frac{1}{4}$

$n.\frac{1}{4} = 4$

$n = 16$

Hence parameters of binomial distribution are B ($n = 16$, $p = \frac{1}{4}$)

$P(X \geq 2) = 1 - P(X = 0) - P(X = 1)$

$P(X = 0) = {}^{16}C_0 \left(\frac{1}{4}\right)^0 \left(\frac{3}{4}\right)^{16} = \left(\frac{3}{4}\right)^{16} = 0.010023$

$P(X = 1) = {}^{16}C_1 \left(\frac{1}{4}\right)^1 \left(\frac{3}{4}\right)^{16-1} = 16 \times \left(\frac{1}{4}\right)^1 \left(\frac{3}{4}\right)^{16-1} = 0.053454$

$P(X \geq 2) = 1 - 0.010023 - 0.053454$

$= 0.936524$

Mode: $(n+1).p = (16+1) \times \frac{1}{4} = \frac{17}{4} = 4.25$

Since 4.25 is in fraction and its integer part is 4, hence mode of distribution is 4.

Example 5.3: Mean of binomial distribution is 3 and variance is 4. Find parameter of the distribution

Solution: Mean $= np = 3$ ..... (i)

Variance $= npq = 4$ ..... (ii)

Divide ii by i

$$\frac{npq}{np} = \frac{4}{3}$$

$q = \frac{4}{3}$, which is impossible because q lies between 0 to 1. It does not follow binomial distribution. (here mean is smaller than variance)

Example 5.4: Mean of binomial distribution is 4 and third central moment $\mu_3 = 0.48$. Find parameter of binomial distribution, mode, variance,

Solution:

Mean $np = 4$

$\mu_3 = npq(q - p) = 0.48$

$$\frac{npq(q-p)}{np} = \frac{0.48}{4}$$

$q(q - p) = 0.12$

put $p = 1 - q$

$q(q - 1 + q) = 0.12$

$2q^2 - q - 0.12 = 0$

$$q = \frac{1 \pm \sqrt{(-1)^2 - 4 \times 2 \times (-0.12)}}{2 \times 2} = \frac{1 \pm \sqrt{1 + 0.96}}{4} = \frac{1 \pm 1.4}{4} = 0.6, -0.1$$

Since probability cannot be negative, $q = -0.1$ is rejected, hence

$q = 0.6$, $p = 0.4$

$0.4n = 4$

$n = 10$

- Mode: (n+1).p = (10+1) × 0.4 =4.4

  Integer part is 4, hence mode of the distribution is 4
- Variance = npq = 10 × 0.4 × 0.6 = 2.4

5.1.3 Additive (or reproductive) property of binomial distribution: If X and Y are two independent binomial variate. X~ B(n_1, p_1) and Y~ B(n_2, p_2), sum of two independent binomial variate does not follow binomial distribution, however if probability of success is same i.e $p_1 = p_2 = p$, then X + Y follows binomial distribution with parameter $n_1 + n_2$ and p i.e X + Y ~ B($n_1 + n_2$, p)

Example 5.5: If X and Y are two independent binomial variate,X~ B(8, $\frac{1}{4}$) and Y~ B(6, $\frac{3}{4}$), what will be distribution of X+Y ?

Solution : Since probability of success is not same for X and Y. X + Y will not follow binomial distribution.

Example 5.6: If X and Y are two independent binomial variate, X~ B(8, $\frac{1}{3}$) and Y~ B(6, $\frac{1}{3}$), what will be distribution of X+Y? Find mean, mode, variance, coefficient of skewness, coefficient of kurtosis and P(X+Y = 2)

Solution: Since probability of success is same for both variate X and Y. X+Y will follow binomial distribution with parameters n= 8 + 6 = 14 and p = $\frac{1}{3}$

Mean = $14 \times \frac{1}{3} = \frac{14}{3} = 4.67$

Mode = $(14+1) \times \frac{1}{3} = 5$, since (n + 1)p is an integer, distribution is bimodal, two modes are $(5-1)$ and 5 i.e. 4 and 5 are two modes.

p = $\frac{1}{3}$, q = $\frac{2}{3}$

Variance = $14 \times \frac{1}{3} \times \frac{2}{3} = \frac{28}{9} = 3.11$

$P(X + Y = 2) = {}^{14}C_2 \left(\frac{1}{3}\right)^2 \left(\frac{2}{3}\right)^{14-2} = 91 \times 0.000856 = 0.07793$

5.1.4 Recurrence relation for fitting the probabilities of binomial distribution: Suppose X follows binomial distribution with parameters n and p, probability mass function of random variable X is

$$P(X = x) = \binom{n}{x} p^x q^{n-x} \qquad (i)$$

$P(X = x + 1) = \binom{n}{x+1} p^{x+1} q^{n-x-1}$ (ii)

Divide (ii) by (i)

$$\frac{P(X = x+1)}{P(X = x)} = \frac{\binom{n}{x+1} p^{x+1} q^{n-x-1}}{\binom{n}{x} p^{x} q^{n-x}} = \frac{\frac{n!}{(x+1)!(n-x-1)!}}{\frac{n!}{(x)!(n-x)!}} \times \frac{p^{x+1-x}}{q^{n-x-n+x+1}}$$

$$= \frac{n!}{(x+1)!(n-x-1)!} \times \frac{(x)!(n-x)!}{n!} \times \frac{p}{q} = \frac{n!}{(x+1)(x)!(n-x-1)!} \times \frac{(x)!(n-x)(n-x-1)!}{n!} \times \frac{p}{q}$$

$$= \frac{(n-x)}{(x+1)} \frac{p}{q}$$

where ${}^{n}c_{x+1} = \binom{n}{x+1} = \frac{n!}{(x+1)!(n-x-1)!}$ ${}^{n}c_{x} = \binom{n}{x} = \frac{n!}{(x)!(n-x)!}$

$P(X = x+1) = \frac{(n-x)}{(x+1)} \frac{p}{q} P(X = x)$ and $P(X=0) = q^n$

Here only P(X = 0) is to be computed and rest probabilities X = 1, 2, …, n is obtained by using recurrence relation.

Example 5.7: 150 sets of seeds were kept for germination. In each set there were 7 seeds. Number of seeds germinated in each set were recorded. Fit binomial distribution to the number of seeds germinated.

X	0	1	2	3	4	5	6	7
f	7	6	19	35	30	23	27	3

Solution:

X	**f**	**fx**	**(n-x)**	$\frac{n-x}{x+1}$	$\frac{n-x}{x+1}\frac{p}{q}$	**p**	**Expected frequency**
0	7	0	7	7.00	8.22	0.00436	0.65 ≅ 1
1	6	6	6	3.00	3.52	0.03580	5.37 ≅ 5
2	19	38	5	1.67	1.96	0.12610	18.92 ≅ 19
3	35	105	4	1.00	1.17	0.24716	37.01 ≅ 37
4	30	120	3	0.60	0.70	0.28918	43.45 ≅ 43
5	23	115	2	0.33	0.39	0.20242	30.61 ≅ 31
6	27	162	1	0.14	0.17	0.07894	11.98 ≅ 12

7	3	21				0.01342	2.01 ≅ 2
Total	150	567				1.00000	150

Mean $= \frac{567}{150} = 3.78$

$np = 3.78$

$p = \frac{3.78}{7} = 0.54$

$q = 1 - 0.54 = 0.46$

$\frac{p}{q} = \frac{0.54}{0.46} = 1.173913$

$P(X=0) = q^n = 0.46^7 = 0.004358$

$E(X) = N \times p$

Example 5.8: If for binomial distribution, $\beta_1 = \frac{1}{15}$, $\beta_2 = \frac{89}{30}$. Find parameter of binomial distribution, mean, mode and variance.

Solution:

$$\beta_1 = \frac{(q-p)^2}{npq} = \frac{1}{15} \quad \text{(i)}$$

$$\beta_2 = 3 + \frac{1-6pq}{npq} = \frac{89}{30}.$$

$$\frac{1-6pq}{npq} = \frac{89}{30} - 3 = -\frac{1}{30} \quad \text{(ii)}$$

Dividing (i) by (ii)

$$\frac{\frac{(q-p)^2}{npq}}{\frac{1-6pq}{npq}} = \frac{\frac{1}{15}}{-\frac{1}{30}}$$

$$\frac{(1-2p)^2}{(1-6p+6p^2)} = -2 \qquad \text{since } (p = 1-q)$$

$1+4p^2 - 4p = -2 + 12p - 12p^2$

$16p^2 - 16p + 3 = 0$

$16p^2 - 12p - 4p + 3 = 0$

$4p(4p-3) - 1(4p-3) = 0$

$(4p-3)(4p-1) = 0$

$4p = 3$ or $4p = 1$

Case i) when $p = \frac{3}{4}$, $q = \frac{1}{4}$

put value of p and q in (i)

$\frac{\left(\frac{1}{4}-\frac{3}{4}\right)^2}{n\times\frac{3}{4}\times\frac{1}{4}} = \frac{1}{15}$

$\frac{\frac{4}{16}}{\frac{3n}{16}} = \frac{1}{15}$

n = 20

mean = $20 \times \frac{3}{4} = 15$

mode = $(20 + 1) \times \frac{3}{4} = 15.75$,

it is in fraction, value of mode is 15

Variance = $20 \times \frac{3}{4} \times \frac{1}{4} = 3.75$

Case ii) when $p = \frac{1}{4}$, $q = \frac{3}{4}$

Variance = $20 \times \frac{1}{4} \times \frac{3}{4} = 3.75$

put value of p and q in (i)

$\frac{\left(\frac{3}{4}-\frac{1}{4}\right)^2}{n\times\frac{1}{4}\times\frac{3}{4}} = \frac{1}{15}$

$\frac{\frac{4}{16}}{\frac{3n}{16}} = \frac{1}{15}$

n = 20

mean = $20 \times \frac{1}{4} = 5$

mode = $(20 + 1) \times \frac{1}{4} = 5.75$,

it is in fraction, value of mode is 5

5.4 Normal Distribution:

Binomial distribution and Poisson distributions reduces to normal distribution:

In case of binomial distribution,

- Coefficient of skewness (γ_1) $= \frac{q-p}{\sqrt{npq}}$

  Binomial distribution reduces to normal distribution when

  i) Number of trial is very large i.e. $n \to \infty$

  ii) $p \approx q$

In case of Poisson distribution

- Coefficient of skewness (γ_1) $= \frac{1}{\sqrt{\lambda}}$

  Poisson distribution reduces to normal distribution when $\lambda \to \infty$

A random variable X is having normal distribution with parameter mean = μ and variance = σ^2 and its probability density function is given by

$$f(X: \mu, \sigma^2) = \frac{1}{\sqrt{2\pi}\,\sigma} \exp\left\{-\frac{1}{2}\left(\frac{X-\mu}{\sigma}\right)^2\right\} \qquad -\infty < X < +\infty\,, -\infty < \mu < +\infty\,,\ \sigma > 0$$

5.4.1 Standard normal variate: To find the probability between two points in a normal distribution, random variable X is converted into standard normal variate Z, where

$$Z = \frac{X-\mu}{\sigma}$$

If $X \sim N(\mu, \sigma^2)$, then $Z \sim N(0, 1)$

Proof: $E(Z) = E\left(\frac{X-\mu}{\sigma}\right) = \frac{1}{\sigma}E(X - \mu)$

$$= \frac{1}{\sigma}\ (E(X) - E(\mu))$$

$$= \frac{1}{\sigma}\ (\mu - \mu) = 0$$

$$V(Z) = V\left(\frac{X-\mu}{\sigma}\right)$$

$$= \frac{1}{\sigma^2} V(X - \mu)$$

$$= \frac{1}{\sigma^2} [V(X) + V(\mu)]$$

$$= \frac{1}{\sigma^2}(\sigma^2+0) = 1$$

Standard normal table is available to obtain the probability between 0 to z. When X = μ, value of Z = 0, when X is less than mean, Z is negative and when X is greater than mean , Z is positive. The curve is symmetrical at mean, hence 0.5 probability is towards the right hand side and 0.5 probability is towards the left hand side of the curve.

Note: To find probabilities between two points

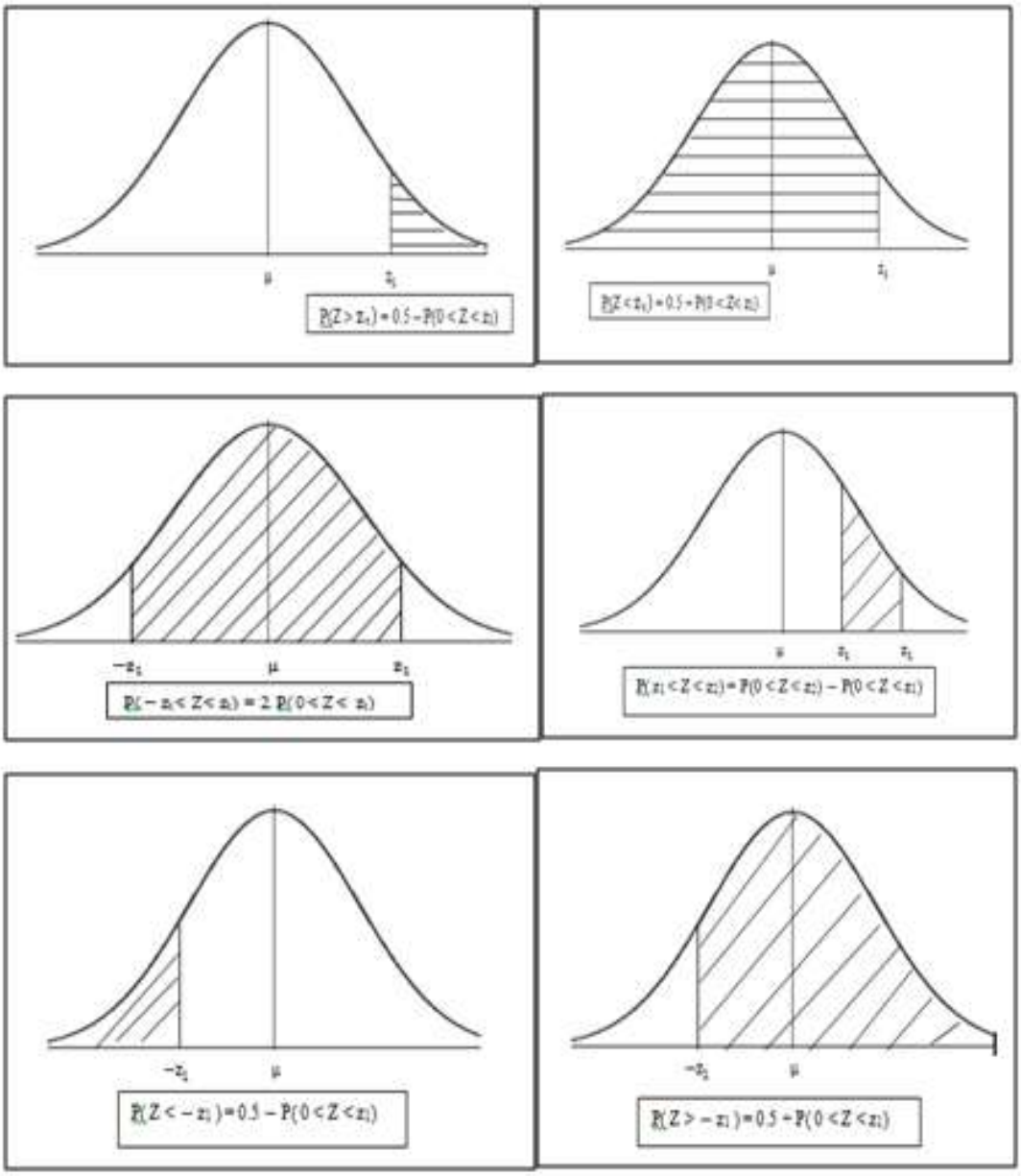
i) $P(Z > z_1) = 0.5 - P(0 < Z < z_1)$

ii) $P(Z < z_1) = 0.5 + P(0 < Z < z_1)$

iii) $P(-z_1 < Z < z_1) = 2.\ P(0 < Z < z_1)$

iv) $P(z_1 < Z < z_2) = P(0 < Z < z_2) - P(0 < Z < z_1)$

v) $P(Z < -z_1) = 0.5 - P(0 < Z < z_1) P(Z > -z_1) = 0.5 + P(0 < Z < z_1)$



5.4.2 Chief characteristics of normal distribution:

i) The curve is bell shaped and symmetrical at mean

ii) Mean, median and mode are same.

iii) All odd order moments are 0

iv) As X increases f(x) decreases rapidly

v) Since f(x) represents probability, no portion of the curve lies below X axis

vi) Quartile deviation, mean deviation and standard deviation are in the ratio of 10:12:15

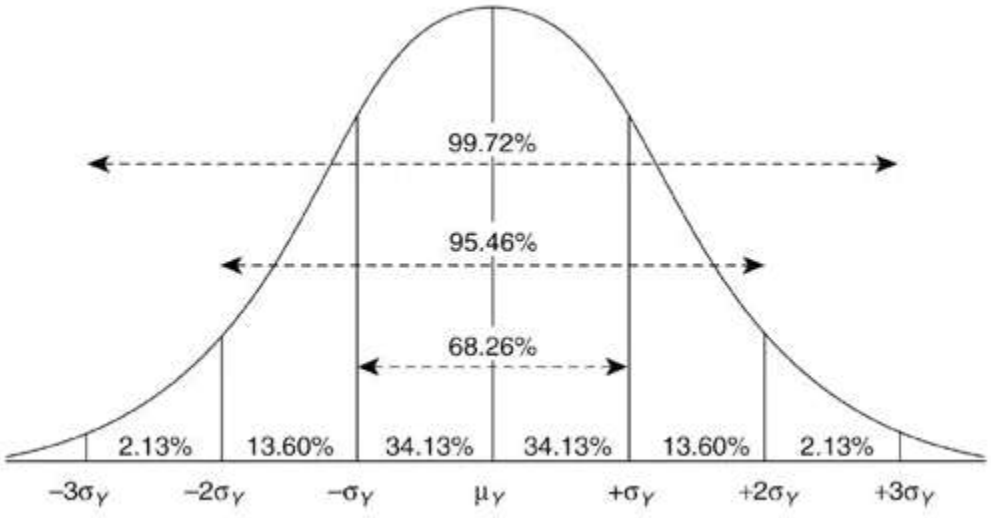
vii) Linear combination of independent normal variate is also normally distributed

viii) Area properties

a) $P(\mu - \sigma < X < \mu + \sigma) = 0.6826$

b) $P(\mu - 2\sigma < X < \mu + 2\sigma) = 0.9544$

c) $P(\mu - 3\sigma < X < \mu + 3\sigma) = 0.9973$



Example 5.21: If X is normally distributed with mean = 50 and variance = 16 find the probability that

i) $P(X > 54)$

ii) $P(X < 54)$

iii) $P(46 < X < 54)$

iv) $P(X < 46)$

v) $P(X > 46)$

vi) $P(54 < X < 58)$

Solution: $X \sim N(\mu = 50, \sigma^2 = 16), \sigma = 4$

i) when $X = 54$

$$Z = \frac{54 - 50}{4} = 1$$

$P(X > 54) = P(Z > 1)$

$= 0.5 - P(0 < Z < 1)$

$= 0.5 - 0.3413$

= 0.1587

ii) $P(X < 54) = P(Z < 1)$

$= 0.5 + P(0 < Z < 1)$

$= 0.5 + 0.3413 = 0.8413$

iii) $P(46 < X < 54)$

When $X = 46$

$Z = \frac{46 - 50}{4} = -1$

$P(46 < X < 54) = P(-1 < Z < 1)$

$= P(-1 < Z < 0) + P(0 < Z < 1)$

$= 2.\ P(0 < Z < 1)$ (Since the curve is symmetrical)

$= 2 \times 0.3413 = 0.6826$

iv) $P(X < 46) = P(Z < -1) = 0.5 - P(-1 < Z < 0)$

$= 0.5 - 0.3413 = 0.1587$

v) $P(X > 46) = P(Z > -1) = 0.5 + P(-1 < Z < 0)$

$= 0.5 + 0.3413 = 0.8413$

vi) $P(54 < X < 58)$

When $X = 58$

$Z = \frac{58 - 50}{4} = 2$

$P(54 < X < 58) = P(1 < Z < 2)$

$= P(0 < Z < 2) - P(0 < Z < 1)$

$= 0.4772 - 0.3413 = 0.1259$

Example 5.22: Weight of fruit is normally distributed with mean weight 60 g and variance 9. Fruits whose weight lies between 57 to 66 g are sold @ Rs 8 per fruit, fruit whose weight is < 57 g are sold @ Rs 6 per fruit and rest are sold @ Rs 10 per fruit. Find total amount collected if in all there are 10000 fruits.

Solution: $X \sim N(\mu = 60, \sigma^2 = 9)$, $\sigma = 3$

When $X = 57$

$Z = \frac{57 - 60}{3} = -1$

When $X = 66$

$Z = \frac{66 - 60}{3} = 2$

$P(57 < X < 66) = P(-1 < Z < 2)$

$= P(-1 < Z < 0) + P(0 < Z < 2)$

$= 0.3413 + 0.4772 = 0.8185$

$P(X < 57) = P(Z < -1) = 0.5 - P(0 < Z < 1) = 0.5 - 0.3413 = 0.1587$

Probability of rest fruits = 1 – 0.8185 – 0.1587 = 0.0228

Total fruits = 10000

Number of fruits whose weight lies between 57 to 66 = 10000 × 0.8185 = 8185

Number of fruits whose weight is less than 57 =10000 × 0.1587 = 1587

Rest fruits = 10000 × 0.0228 = 228

Weight of fruit (g)	Number of fruits	Price (Rs)	Amount (Rs)
57 to 66	8185	8	65480
< 57	1587	6	9522
Rest fruits	228	10	2280
Total	10000		77282

Total amount collected = 77282

5.4.3 Linear combination of independent normal variate: If X and Y are two independent normal variate. $X \sim N(\mu_1, \sigma_1^2)$ and $Y \sim N(\mu_2, \sigma_{2,}^2)$, then

$E(aX + bY) = aE(X) + bE(Y)$

$V(aX + bY) = a^2V(X) + b^2V(Y)$

If $a = 1, b = 1$

$E(X + Y) = E(X) + E(Y) = \mu_1 + \mu_2$

$V(X + Y) = V(X) + V(Y) = \sigma_1^2 + \sigma_2^2$

If $a = 1, b = -1$

$E(X - Y) = E(X) - E(Y) = \mu_1 - \mu_2$

$V(X - Y) = V(X) + V(Y) = \sigma_1^2 + \sigma_2^2$

5.4.4 Additive properties of normal distribution: If X and Y are two independent normal variate and $X \sim N(\mu_1, \sigma_1^2)$ and $Y \sim N(\mu_2, \sigma_{2,}^2)$, then sum as well as difference of two independent normal variate are also normally distributed i.e.

$(X + Y) \sim N(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$

$(X - Y) \sim N(\mu_1 - \mu_2, \sigma_1^2 + \sigma_2^2)$

Example 5.23: If X and Y are two independent normal variate, X ~N(20, 9) and Y~N(10,4), what will be the distribution of i) 2X + 3Y ii) 2X – 3Y iii) X – Y iv) X + Y

i) 2X + 3Y

$E(2X + 3Y) = 2E(X) + 3E(Y) = 2 \times 20 + 3 \times 10 = 70$

$V(2X + 3Y) = 4V(X) + 9V(Y) = 4 \times 9 + 9 \times 4 = 72$

$2X + 3Y \sim N(70, 72)$

ii) 2X – 3Y

$E(2X - 3Y) = 2E(X) - 3E(Y) = 2 \times 20 - 3 \times 10 = 10$

$V(2X - 3Y) = 4V(X) + 9V(Y) = 4 \times 9 + 9 \times 4 = 72$

$2X - 3Y \sim N(10,72)$

iii) $X - Y$

$E(X - Y) = E(X) - E(Y) = 20 - 10 = 10$

$V(X - Y) = V(X) + V(Y) = 9 + 4 = 13$

$X - Y \sim N(10, 13)$

iv) X + Y

$E(X + Y) = E(X) + E(Y) = 20 + 10 = 30$

$V(X - Y) = V(X) + V(Y) = 9 + 4 = 13$

$X + Y \sim N(30, 13)$

Example 5.24: Fruit yield of plant is normally distributed with mean number of fruits 600 and variance 1600. If there are 1000 plants in an orchard, how many plants have i) mean fruit yield less than 700 ii) between 650 to 700 iii) more than 650

Solution: $X \sim N(600, 1600)$ $\sigma = 40$

When X = 700

$Z = \frac{700 - 600}{40} = 2.5$

i) $P(X < 700) = P(Z < 2.50)$

$= 0.5 + P(0 < Z < 2.5) = 0.5 + 0.4938 = 0.9938$

Number of plants having fruit yield less than 700 = 1000 × 0.9938 = 994

ii) $P(650 < X < 700)$

When X = 650

$Z = \frac{650 - 600}{40} = 1.25$

$P(650 < X < 700) = P(1.25 < Z < 2.50) = P(0 < Z < 2.50) - P(0 < Z < 1.25)$

$= 0.4938 - 0.3944 = 0.0994$

Number of plants having fruit yield between 650 to 700 = $1000 \times 0.0994 = 99$

iii) $P(X > 650) = P(Z > 1.25) = 0.5 - P(0 < Z < 1.25) = 0.5 - 0.3944 = 0.1056$

Number of plants having fruit yield more than $650 = 1000 \times 0.1056 = 106$

Example 5.25: 1000 students are studying the course of statistics in the university. Probability that a student require Statistics book from the library is 0.45. How many books should be available in the library such that probability of getting issue of a book from library is atleast 0.8

Solution: Here n = 1000, p = 0.45 q = 0.55, n is large and $p \approx q$, binomial distribution reduces to normal distribution

Mean = np = $1000 \times 0.45 = 15.73$

Variance = npq = $1000 \times 0.45 \times 0.55 = 247.5$

Standard Deviation $\sigma = 15.73$

From the standard normal table area from $-\infty$ to + 0.84 is 0.80 i.e from 0.84 to $+\infty$ is 0.20

Let X represents number of books available in the library

$\frac{X - \mu}{\sigma} > 0.84$

$\frac{X - 450}{15.73} > 0.84$

$X - 450 > 0.84 \times 450$

$X > 13.22 + 450$

$X > 463.22$

So library must have atleast 464 books in the library

5.4.5 Fitting of Normal distribution: Suppose X is normally distributed with mean = μ and variance = σ^2 . We assume that first class is less than upper limit of that class and last class as more than lower limit of that class, then convert all upper limit into standard normal variate Z.

Example 5.25: Height (feet) of 1000 trees was measured and following is the distribution of height of trees.

Class	0-10	10-20	20-30	30-40	40-50	50-60	60-70	70-80	80-90	90-100	Total
f	2	8	38	96	145	272	220	172	44	3	1000

Solution:

Class (Height)	**f**	**MV (X)**	**d = $\frac{X-45}{10}$**	**fd**	**fd^2**	**Z**	**Probability from 0 to z**	**p**	**Expected Frequency**
0-10	2	5	-4	-8	32	-3.07	0.4989	0.0011	1
10-20	8	15	-3	-24	72	-2.42	0.4922	0.0067	7
20-30	38	25	-2	-76	152	-1.77	0.4616	0.0306	31
30-40	96	35	-1	-96	96	-1.12	0.3686	0.0930	93
40-50	145	45	0	0	0	-0.47	0.1808	0.1878	188
50-60	272	55	1	272	272	0.19	0.0754	0.2562	256
60-70	220	65	2	440	880	0.84	0.2996	0.2242	224
70-80	172	75	3	516	1548	1.49	0.4319	0.1323	132
80-90	44	85	4	176	704	2.14	0.4838	0.0519	52
90-100	3	95	5	15	75			0.0162	16
Total	1000			1215	3831			1	1000

Mean $= 45 + 10 \times \frac{1215}{1000} = 45 + 12.15 = 57.15$

Standard Deviation $= \sqrt{\frac{10^2}{1000}[3831 - \frac{1215^2}{1000}]} = \sqrt{\frac{10^2}{1000}[3831 - 1476.225]}$

$= \sqrt{\frac{10^2}{1000} \times 2354.775} = \sqrt{235.48} = 15.35$

$Z = \frac{10 - 57.15}{15.35} = -3.07$

$P(0 < X < 10) = P(X < 10) = P(Z < -3.07) = 0.5 - 0.4989 = 0.0011$

$Z = \frac{20 - 57.15}{15.35} = -2.42$

P (10 < X < 20) = P(−3.07 < Z < −2.42) = 0.4989 – 0.4922 = 0.0067

$Z = \frac{30 - 57.15}{15.35} = -1.77$

P(20 < X < 30) = P(−2.42 < Z < − 1.77) = 0.4922 – 0.4616 = 0.0306

$Z = \frac{40 - 57.15}{15.35} = -1.12$

P(30 < X < 40) = P(−1.77 < Z < − 1.12) = 0.4616 − 0.3686 = 0.0930

$Z = \frac{50 - 57.15}{15.35} = -0.47$

P(40 < X < 50) = P(−1.12 < Z < − 0.47) = 0.3686 – 0.1808 = 0.1878

$Z = \frac{60 - 57.15}{15.35} = 0.19$

P(50 < X < 60) = P(− 0.47 < Z < 0.19) = P(−0.47 < Z < 0) + P(0 < Z < 0.19)

= 0.1808 + 0.0754 = 0.2562

$Z = \frac{70 - 57.15}{15.35} = 0.84$

P(60 < X < 70) = P(0.19 < Z < 0.84) = 0.2996 – 0.0754 = 0.2242

$Z = \frac{80 - 57.15}{15.35} = 1.49$

P(70 < X < 80) = P(0.84 < Z < 1.49) = 0.4319 − 0.2996 = 0.1323

$Z = \frac{90 - 57.15}{15.35} = 2.14$

P(80 < X < 90) = P(1.49 < Z < 2.14) = 0.4838 − 0.4319- = 0.0519

P (90 < X < 1000) = P (X > 90) = P(Z > 2.14) = 0.5 − 0.4838 = 0.0162

5.4.6 Importance of Normal distribution:

i) Most of the distribution occurring in practice i.e. binomial, Poisson and other distributions can be approximated by normal distribution

ii) Most of the sampling distributions i.e. t, F and $\chi 2$ approaches to normal distribution for large sample

iii) The distribution of sample statistic i.e. sample mean, variance approaches to normal distribution for large sample.

iv) Entire theory of small sample test i.e. t, F and $\chi 2$ is based on the fundamental assumption that parent population from which the sample have been drawn is normally distributed.

v) Normal distribution has wide applications in quality control

vi) If variable X is not normally distributed, then by suitable transformation, the variable reduces to normal distribution. If variable X is skewed distribution, then square root of X is normally distributed.

In case of percentage data, where percentage is calculated from the count, data can be easily transformed to normal data by using suitable transformations.

a) If percentage lies between 30 -70, data is normally distributed, hence no transformation is used

b) If percentage values lie between 0 to 30 or 70 to 100, square root transformation is used.

c) If both conditions are not fulfilled, angular transformation is used.